

Shuffle

```
/**
 * Write a static, generic method called 'shuffleCollection' that receives a Collection
 * whose elements are restricted to type Number.
 * <p>
 * The method returns a new, immutable collection of the same type as the parameter with the same elements, but in a random order.
 * <p>
 * NOTE: The use of {@link Collections#shuffle} is forbidden!!
 */
final class ShuffleUtil {

    private static final Random RANDOM = new Random(); // Might be useful.

    // TODO

    public static void main(String[] args) {
        System.out.println(shuffleCollection(List.of(1, 2, 3, 4, 5, 6, 7, 8, 9))); // [9, 3, 5, 6, 1, 8, 2, 4, 7]
    }
}
```

Shuffle: Lösung

```
final class ShuffleUtil {  
  
    private static final Random RANDOM = new Random();  
  
    static <T extends Number> Collection<T> shuffleCollection(Collection<T> collection) {  
        List<T> result = new LinkedList<>(collection);  
  
        for (int i = result.size() - 1; i > 0; i--) {  
            int j = RANDOM.nextInt(i + 1);  
            swap(result, i, j);  
        }  
  
        return Collections.unmodifiableList(result);  
    }  
  
    private static <T> void swap(List<T> list, int i, int j) {  
        T temp = list.get(i);  
        list.set(i, list.get(j));  
        list.set(j, temp);  
    }  
}
```